Agentic AI for teaching

What we've been doing, what it costs, and where the judgment sits

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Today's plan

1:00 — Emily

- **Basics recap:** chatbots vs. agents, and what an agent can touch
- **Where AI can help w/ teaching prep:** story time
- **Grading and student data:** Principles and a pipeline suggestion

1:45 — Erkmen

- **What the evidence says** about AI in teaching so far
- **How others are using it**
- **Live demo:** an interactive dashboard your students can use

Chatbot vs. agent

A chatbot

- You type a message, it replies
- You copy the output somewhere
- It has no idea what's on your computer
- Each turn starts fresh

An agent

- **Reads your actual files** — slides, syllabus, problem sets, code
- **Runs commands** — Stata, Python, LaTeX, git
- **Remembers across turns** — standing instructions, course context
- Asks permission before doing anything risky

What it can touch — and how you widen it

1. Default

Your working directory only.

Asks before each action. Where you start, and where most work happens.

2. Widen

Grant more as trust grows.

More folders, standing permissions for actions you've vetted — loosened deliberately.

3. Sandbox / VM

Isolated environment.

For big, autonomous jobs — it runs freely in a space that *can't* touch the rest of your machine.

This matters for Part 2: **you decide what the agent can see.** Student data is the case where that decision has rules attached.

Part 1

Where it fits in teaching prep

What got cheaper, what didn't, and one story about Beamer

Two things it can buy you — 1. Less time

Less time

The obvious one: formatting, converting, "make this deck match that deck," "turn these notes into a problem set."

- The mechanical hour before class that used to be the whole prep
- Getting a draft of *anything* on the page so there's something to react to

Real, and it's why most people start — but on its own it mostly makes a *good-enough* lecture very, very cheap.

Two things it can buy you — 2. More range

More range

Things I used to *not do* because the fixed cost was too high in a busy semester:

- Bring in the paper I read last week
- A plug-and-play demo students can open in a browser
- One more worked example, at 11pm, the night before

It makes a **better** lecture **feasible**, not just a faster one.

A story about Beamer (2015)

Third time teaching statistics. Feeling like *quite* the expert educator. I decided to convert my ugly PowerPoint slides to Beamer, for reasons that seemed extremely important at the time.

Every week, on the couch: convert next week's lecture, slide by slide.

Not a great use of time on its face, but it was the right balance of tedious and engaging. I sat with every bullet point and asked what it was doing there.

Best semester I'd taught yet.

The only change students could see: no more PowerPoint.

It wasn't about the slides

It was the **line-by-line pass** the silly task forced me into.

Complement, not substitute

Econometrics, this past year — what the workflow actually looked like:

The sequence

1. Write one giant prompt (Scott Cunningham's recommendation): standardize the format, add examples, tighten the arc
2. Let the agent work through the whole deck
3. **Then don't trust it.** Sit down and read the revised slides line by line, extremely slowly
4. ...which produced *more* changes than step 2 did — new activities, reordering, cuts

✔ When it went well

A fresh pair of eyes on a deck: "Does this work? What's missing here?"
It made everything pretty *and* also a little bit broken

✖ When I skipped step 3

It tried to sneak in a line that *not* violating the zero-conditional-mean assumption implies causality. 😱

Same story as coding: the returns come from the verification pass, not from the generation.

Three workflows that worked — and two that didn't

Worked

1. **Add new content.** Hand it a paper → draft slides in my format → I edit
2. **Audit a deck.** "Where are the holes? Does this hang together? What would a student get lost on?"
3. **Build a demo and embed it.** An interactive supply-and-demand explorer, a CLT visual — one HTML file, no login. *(Erkmen will show this.)*

If a workflow works, save it — as a prompt file or a skill — and reuse it next semester.

Didn't

- Two hours arguing with Claude about the appearance of call-out boxes. (Students can't learn from ugly slides, apparently.)
- A JavaScript demo I was too excited about, and then lost my mouse in, mid-lecture

Every concern we have about students shortcutting their learning applies to us shortcutting our prep.

Choose your headwinds

If you want to set a personal best, you avoid the headwind. If you want to get *stronger*, a steady headwind is your friend — it keeps you pushing the whole way, no coasting.

Fast

If the goal is to go as fast as you can, these tools will do that — used well.

Better

If the goal is to be a better teacher, keep some friction — and **you get to pick which**.

Which frictions force the line-by-line pass? Which ones are just tedium? I don't have the general answer — but that's the design question.

Part 2

Grading and student data

The UVM rules in 30 seconds, and a rubric an agent can actually grade

The 30-second version (UVM edition)

FERPA applies the moment identifiable student work or grades go into an AI tool. Two clean paths.

✔ Path 1: UVM-approved tool

Microsoft Copilot, logged in with your NetID — approved for high-risk data including student academic records.

✔ Path 2: De-identify first

Strip the identifiers so it's no longer an education record. Then use whatever tool you like — including the good one you pay for yourself.

Not this path

Personal ChatGPT, personal Claude, Gemini: **not approved for UVM data**. (Yet!!) A named essay or a gradebook pasted into one is a disclosure the university hasn't authorized.

What counts as identifiable student data

FERPA covers. . .

- Grades, transcripts, class lists, student work
- Records **held by a vendor on your behalf** (they stay protected after handoff)
- **Direct** identifiers (name, ID, email)
- **Indirect** identifiers (DOB, writing style, topic)

The test: could a reasonable person in your community identify the student? If yes, it's identifiable.

The mosaic effect

In a **small seminar**, an "anonymized" essay *could* still be re-identifiable from:

- topic choice
- writing style
- facts you already know about the student

De-identification means addressing **combinations** — not just deleting the name.

A *student* pasting their *own* work into a consumer tool is their choice. The obligation runs to **you and the institution**.

Concrete classroom scenarios

Scenario	Verdict	Why
Named essay into personal ChatGPT for feedback	✗	Unauthorized disclosure
Same essay, **fully de-identified**, low re-ID risk, any tool	⚠	Defensible — your judgment
Same essay, in **Copilot with your NetID**	✓	UVM-approved for student data
Upload the class gradebook to summarize performance	✗	Grades are PII (unless Path 1)
Course chatbot **trained on** student submissions	✗	Needs approval + agreement
AI to draft a rubric, lecture, or problem set (no student data)	✓	No education records involved

The dividing line is almost always: **identifiable student data + a tool the university doesn't control.** Small seminars make "de-identified" harder than it sounds — topic and writing style re-identify.

A rubric an agent can grade

Before you write it

1. **Cursory pass first.** Skim everything, list the edge cases, write the rules *before* grading — raw Stata output? null results? missing heading vs. missing section?
2. **Show it your own feedback.** Real comments you wrote to real students, plus your grades on the earlier assignments. This is the difference between generic feedback and feedback that sounds like you — and knows what you already told them.

In the rubric itself

3. **"Full points when. . . "** — criterion-based, not norm-referenced. "4 = exceptional" produces hedged 4.8s.
4. **Gates over sliders.** Word count, all questions present, attribution statement: binary and stable. Six-point scales drift.
5. **Split check from judge.** Checks are where AI is reliable. Judgment stays yours.

Before you trust it

6. **Do a test grade.** Run it on something you've already scored yourself — last term's paper, or a fake one with flaws you planted. That's how you find out where it drifts, before it touches a real student.
7. **Plan the second pass.** Budget it. You will recalibrate.

One rubric line, three different questions

"Important literature discussed and linked to the RQ; at least four peer-reviewed sources distinct from the intro." **4 pts**

↓ one line, but three different kinds of question ↓

✔ Countable

Four peer-reviewed, distinct from the intro's two. A working paper isn't one.

Make it a gate. Same answer every time, from anyone.

Checkable

Is each source actually discussed, not just cited? Tied back to the question? Does it say what's still unknown?

Hand it to the agent. This is what it's reliably good at.

Judgment

Synthesis, or summary?

Keep it. And watch it — this is the line every grader inflated.

What I asked them for

ECON3500: Econometrics and Applications   Syllabus Schedule Content **Assignments**

Research Paper: Research Proposal

Due by **1:15 PM** on Thursday, April 9, 2026

Objective

The goal of this submission is for you to translate your research idea and data set into the outline of a workable paper. You can think of your research proposal as "baby paper": a summary of what your question is, why it matters, and how you intend to solve it.

For this assignment, **some basic data work is necessary.** However, you may find it helpful to explore some analyses to get a better sense of what empirical specifications are feasible.

Components

While our first two assignments were fairly informal, this is a formal paper. That means that I will pay close attention to not only what ideas you present, but also *how* you present them.

Your proposal should be **at least 1200 words** (excluding references) and include the following components:

1. **Introduction** that contains (1) a clearly stated research question. What hypotheses are you testing? (2) Motivation — why is this important/interesting? Include at least 2 sources.
2. **Literature review**: discussion of related literature and how your paper fits in. Include at least 4 peer-reviewed sources that are *distinct from* your introduction sources (i.e., at least 6 unique sources total across the introduction and literature review).

What makes it gradeable

- **Countable things.** 1,200 words. 2 sources in the intro, 4 more in the lit review, 6 unique overall.
- **Named components.** Eight of them, each with its own rubric line — so "missing" is a fact, not an opinion.
- **38 points**, five levels per criterion, "full points when. . . " written out.

What came in — except this one is fake

Paid Family Leave and Mothers' Labor Force Participation: Evidence from State Policies, 2010–2023

Andy Dwyer

ECON 3500 — Research Proposal — April 9, 2026

1. Introduction

How does paid family leave affect mothers? Since California introduced the first state paid family leave (PFL) program in 2004, eight states have adopted programs that replace part of a worker's wages for several weeks after the birth of a child. Supporters argue that paid leave keeps mothers attached to their jobs, while critics worry that it raises costs for employers and could reduce hiring of women of childbearing age. In this paper I ask whether state PFL laws increase the labor force participation of mothers with young children.

This question matters because women's labor force participation in the United States has stalled since the 1990s and has fallen behind most other rich countries. Blau and Kahn (2013) find that almost a third of the gap between the U.S. and other OECD countries can be explained by the absence of family-friendly policies such as paid leave. Goldin (2014) argues that the "last chapter" of gender convergence depends on the flexibility of jobs, and paid leave is one of the main policies that can change how costly it is for a mother to stay employed. The *family gap* in pay documented by Waldfogel (1998) also suggests that the years right after a birth are when women's careers are most vulnerable.

To answer this question I use the Current Population Survey Annual Social and Economic Supplement (CPS ASEC) from 2010 to 2023, obtained through IPUMS. I compare mothers of children under five in states with a PFL program in effect to mothers in states without one, controlling for individual characteristics and for state and year fixed effects. This paper will show the causal effect of paid leave on mothers' employment and help policymakers in states that are currently debating PFL legislation.

2. Literature Review

The most studied PFL program is California's. Rossin-Slater, Ruhm, and Waldfogel (2013) use the CPS and a difference-in-differences design to show that California's program roughly doubled leave-taking among mothers of infants and increased weekly hours and wages of mothers of one- to three-year-olds. The effects were largest for less-advantaged mothers, who previously had little access to paid leave.

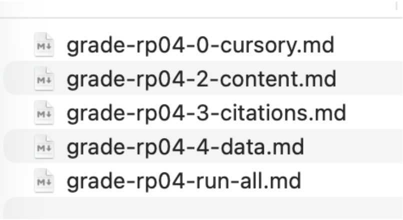
Baum and Ruhm (2016) use the National Longitudinal Survey of Youth 1997 and find that California's PFL raised the probability that mothers were working one year after birth by about

1

Why fake

- I can't put a real student's paper on this screen — so I wrote one. Fictional author, fictional numbers, real citations, a topic unlike anything my S26 class turned in.
- **14 flaws planted on purpose**, each mapped to a rubric line: a lit review that summarizes instead of synthesizes, a working paper counted as peer-reviewed, "this will *prove*" where it should hedge, a table with no group split, no clustering mentioned.
- **Answer key written first**, before any grader saw it. Target: **28.5/38** — deliberately mid, because that's where graders actually differ. (Real class: median 31.25.)

How it actually runs — four passes, one command



One job each

- **cursory** — an overview pass. Skim all 21 first, list the edge cases, write the rules *before* anyone is graded.
- **content** — the assignment and the rubric. The 38 points, and the feedback the student actually reads.
- **citations** — a deliberately mechanical check: do these sources exist, are they peer-reviewed, do the counts hold? *(I care about this more than I can fully justify.)*
- **data** — will the data do what they say it will? Are the summary stats internally consistent? Does the specification match the data described?

Context I hand them: the assignment description, *and* my own feedback and grades on the two earlier assignments — which is what lets the feedback say "you did follow up on the concern I raised in February."

For 21 submissions, I'm honestly not sure it was *faster*. It was more **consistent** — and that's the part I couldn't do by hand at 11pm.

Five graders, one proposal

A fake ECON 3500 proposal with 14 planted flaws. My real Spring 2026 grading prompt. Target: 28.5/38.

Component (pts)	Key	Copilot	Fable	Sonnet	Codex	Opus
Introduction (6)	4.5	6	6	6	6	6
Literature (4)	2	3	**2**	3	**2**	**2**
Data + table (6)	5	6	**5**	6	**5**	**5**
Empirical (10)	7	7.5	8	8	7.5	8
Threats (2)	1.5	2	2	2	2	2
Presentation (10)	8.5	9.5	9.5	9.5	9.5	**8.5**
Total (38)	**28.5**	**34**	**32.5**	**34.5**	**32**	**31.5**
Noticed (of 14)	—	7*	13	8	11	13
Deducted for	—	6*	10	6	9	**12**
Flaws invented	—	0	0	0	0	0

Copilot (UVM NetID) · Fable 5, Sonnet 5, Opus 5 as fresh blind agents · Codex (GPT-5.6-sol, high) by hand.
*Copilot is a floor: only its score breakdown was captured.

What happened

- **Everyone drifted up** — always on the judgment criteria. But not equally: Opus +3, Sonnet +6.
- **The checks held.** Both overclaims, the missing reference, the source count: all five.
- **The gap is "noticed" vs. "deducted."** Sonnet saw 8 and cut for 6. Opus saw 13 and cut for 12.
- **Nobody invented anything.** Zero hallucinated defects, all five.

Read the notes, not the numbers. And one planted flaw **nobody caught** — all five said the

What the feedback read like

Each grader also wrote the student feedback my prompt asks for. Compared to one I wrote in April.

On this paper

- **Codex** wrote the best next steps — and handed the student the exact sentence to use instead of "prove." My length.
- **Fable** was the most complete, and got the structure of my feedback: the name, the quoted sentences, the headers. Three times my length.
- **Sonnet** was fine — but called four summaries "genuinely good synthesis."
- **Nobody** got the pivot. No "That said," no closing on the positive.

But across everything I've graded

- **Codex has been the harsher grader**, and the least likely to pad a score. That part is genuinely useful.
- **But it reflects less.** It applies criteria in odd ways, and the feedback comes out less constructive.
- **And it doesn't get my voice** — consistently the furthest from how I actually write to a student.
- *All of which may change with the next model. This is a snapshot, not a ranking.*

What I'd actually tell you to do

Do

- **Split the rubric.** Countable → gate. Checkable → agent. Judgment → you.
- **Give it your context.** The assignment, and your own past feedback.
- **One job per pass**, and let them disagree — the disagreements are the product.
- **De-identify first.** Mine never sees a name, which is what lets me use whatever model grades best.

Expect

- **It drifts up** — and always on the judgment lines. You just watched five graders do it.
- **One pass is never enough.** Budget the second one.
- **Models differ, and it moves.** What was true in April wasn't true yesterday.
- **The checks hold.** 133 citations verified, none fabricated; the audits caught a crashed log, a 14-line do-file, a blank template.

When it's worth it

- **Worth it** when there are many submissions, many criteria, and real mechanical checking to do.
- **Not worth the plumbing** for three-question essays — Copilot and a good rubric are enough. The rubric is the same file either way.
- **At 21 submissions it wasn't clearly faster.** It was more consistent.

Every correction I've had to make landed on a *judgment* criterion. None landed on a gate or a check.

“Should AI be grading at all?”

A fair question, and one a faculty attendee put to us bluntly at the June webinar. My honest answer, in three parts:

1. What I let it grade

Rubric-gated completion work — did they answer all six questions, at length, thoughtfully. Not the term paper. Not anything where the grade *is* the feedback.

2. It’s an RA, not a referee

Same contract as a TA: it does a first pass, I check what it flags and spot-check what it didn’t, and **the errors are mine either way**. Nothing about that changed.

3. Consistency is a process

My own first pass under-scored 11 of 17 papers; the second pass caught it. The comparison isn’t AI vs. a perfect grader — it’s AI-plus-review vs. me, essay 43 of 60, at 10pm.

What I don’t have a settled answer on: **disclosure.** Do we tell students? What do we tell them? I’d genuinely like to know where this room lands.

Over to Erkmen

My half in one line: it makes a better lecture feasible and a good-enough one cheap · know the UVM rule · write a rubric with gates and checks · let the agent do the checks · you keep the judgment.

Erkmen: what the evidence says, how others are using it, and a live dashboard.